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CUBIC GAUSS SUMS AND COVERS OF SL(2), FOLLOWING S. J. PATTERSON

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0. Introduction.

A Gauss sum is the scalar product of an additive character and a multiplicative character of a finite field. If F_q is a finite field with q elements, χ a character of F_q^*, and ψ a character of the additive group of F_q, set

$$g(\chi,\psi) = \sum_{x \in F_q^*} \chi(x)\ \psi(x).$$

We assume in principle that χ and ψ are nontrivial; the sum $g(\chi,\psi)$ then has absolute value $q^{1/2}$. It is called an n-th order sum if χ has order n. For a $\in F_q^*$, one has

(0.1) $g(\chi,\psi(ax)) = \chi(a)^{-1}\ g(\chi,\psi)$.

In particular, if χ has order n, $g(\chi,\psi)^n$ is independent of ψ.

To each prime number p $\neq$ 2 one can attach the quadratic Gauss sum

$$g_p = \sum_{x \in F_p^*} (x/p)\ \exp(2\pi i\, x/p),$$

where (x/p) is the Legendre symbol : +1 if x is a square modulo p and -1 otherwise. Gauss calculated this sum : it is $\sqrt{p}$ or $i\sqrt{p}$ according as p $\equiv$ 1 or -1 modulo 4.

If, similarly, one wishes to attach an n-th order Gauss sum to a prime ideal $\mathfrak{p}$ of the ring of integers O of a number field F, one must define a multiplicative character, and an additive character of the residue field $O/\mathfrak{p}$.

Suppose that F contains a primitive n-th root of unity, and put $\mu := \mu_n(F)$. If $\mathfrak{p}$ is prime to n, there is an isomorphism $\mu \simeq \mu_n(O/\mathfrak{p})$.

Let t denote the inverse isomorphism, and put q = $N\mathfrak{p}$. We write $(x/\mathfrak{p})_n$, or simply $(x/\mathfrak{p})$, for the character $t(x^{(q-1)/n})$ of $(O/\mathfrak{p})^*$ with values in μ. More generally, if $\mathfrak{a}$ is an ideal of O prime to x, with prime-ideal factorization $\mathfrak{a} = \prod \mathfrak{p}_i^{a_i}$, we write $(x/\mathfrak{a})_n$, or simply $(x/\mathfrak{a})$, for the product $\prod (x/\mathfrak{p}_i)^{a_i}$ (for x prime to $\mathfrak{a}$). For $\mathfrak{a}$ = (a), we also write (x/a). Its relation with the Hilbert symbol (section 2) is the formula

$$(x/a) = \prod_{v|a} (x,a)_v.$$

The reciprocity law for the Hilbert symbol yields a reciprocity law for these symbols.

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The symbol $(x/\mathfrak{a})$ is a character of $(O/\mathfrak{a})^*$ (and already of $(O/\prod \mathfrak{p}_i)^*$) with values in μ. To obtain from it a complex-valued character, it suffices to compose it with an isomorphism ε from μ to the group of n-th roots of unity in C.

We shall denote by e the character $\exp(2\pi i x)$ of R/Z, the induced character on Q/Z, those deduced from it on the Q_p/Z_p through the isomorphism $Q/Z \simeq \oplus Q_p/Z_p$, and those deduced from it by traces : for example, the character $\exp(2\pi i\, Tr_{F/Q}$ x) of F.

There are two methods for constructing additive characters of residue fields.

First method : For an ideal $\mathfrak{p}$ of O, with residue field $O/\mathfrak{p}$ of characteristic p, define the additive character ψ as the composite of the trace $O/\mathfrak{p} \to Z/pZ$ and the character $e(x/p)$ of Z/pZ.

Second method : We spell this out only in the following special case : n = 3 and F is the field $Q(\sqrt{-3})$ of cube roots of unity. Throughout the Introduction, unless expressly stated otherwise, we shall consider only this case. Every ideal $\mathfrak{a}$ prime to 3 has a unique generator a congruent to 1 mod 3, and as additive character of $O/\mathfrak{a}$ we take $e(a^{-1}x)$.

For prime $\mathfrak{p}$, the cubic Gauss sums obtained by using the additive characters from the two methods coincide : for $\mathfrak{p} = (\pi)$, with $\pi \equiv 1$ (3), an easy calculation (cf. (0.1)) shows that one is the product of the other by $(-1/\pi)$ if $\mathfrak{p}$ has degree 2, and by $(\overline{\pi}/\pi)$ if $\mathfrak{p}$ has degree 1. These cube roots of unity are trivial : $(-1/\pi)$ because $(-1)^2 = 1$, and $(\overline{\pi}/\pi)$ because it is real ; indeed, $\overline{(\overline{\pi}/\pi)} = (\pi/\overline{\pi})$ (by transport of structure), while $(\overline{\pi}/\pi) = (\pi/\overline{\pi})$ (by cubic reciprocity).

The second method makes it possible to attach a sum, still called a "Gauss sum," to an ideal $\mathfrak{a}$ that need not be prime. For $\mathfrak{a}$ = (a), with a ≡ 1 (3), put

$$(0.2)\ g(\mathfrak{a}) = \sum_{x \in (O/\mathfrak{a})^*} (x/\mathfrak{a})e(a^{-1}x).$$

One has :

(0.3) If $\mathfrak{a}$ is squarefree, $g(\mathfrak{a}) = (N\mathfrak{a})^{1/2}$. Otherwise, $g(\mathfrak{a}) = 0$. (0.4) Twisted multiplicativity : If $\mathfrak{a} = \mathfrak{a}' \cdot \mathfrak{a}''$, with $\mathfrak{a}'$ and $\mathfrak{a}''$ relatively prime, and with generators a' and a" congruent to 1 mod 3, then

$$g(\mathfrak{a}) = (a'/\mathfrak{a}'')(a''/\mathfrak{a}')\ g(\mathfrak{a}')\ g(\mathfrak{a}'').$$

Cubic reciprocity gives $(a'/\mathfrak{a}'') = (a''/\mathfrak{a}')$. Thus the factor $(a'/\mathfrak{a}'')(a''/\mathfrak{a}')$ may be replaced by $(a'/\mathfrak{a}'')^2$.

For arbitrary F and n, Weil [22] showed that the n-th power of the Gauss sum $g(\mathfrak{p})$, as a function of $\mathfrak{p}$, is an algebraic Hecke character. Notice that this n-th power does not depend on the choice of additive character. If this Hecke character is regarded as known, then, in order to calculate the sum $g(\mathfrak{p})$ itself, one must determine which n-th root of $g(\mathfrak{p})^n$

This is what Gauss does in the quadratic case; and, when n is even, n = 2m, his result reduces the calculation of $g(\mathfrak{p})^m$ to that of an algebraic Hecke character. There are also p-adic results (Stickelberger, recently sharpened by Gross-Koblitz [4]) and, for n = 3,4 and F = $Q(\sqrt[n]{1})$, an expression in terms of a product of division values of elliptic functions (Matthews [13]).

In [19], Patterson proves that the argument $g(\mathfrak{a})/|g(\mathfrak{a})|$ of $g(\mathfrak{a})$, for $\mathfrak{a}$ prime to 3 and squarefree, is equidistributed on the unit circle. He uses Weyl's criterion, and must therefore prove that, for every nonzero integer T,

$$(0.5)\ \Big|\sum_{N\mathfrak{a}\leq x} (g(\mathfrak{a})/|g(\mathfrak{a})|)^T\Big| \leq o(X).$$

The Wiener-Ikehara Tauberian theorem allows such an estimate to be deduced from the holomorphy, for Re(s) ≥ 1, of the Dirichlet series

(0.6) $\sum (g(\mathfrak{a})/|g(\mathfrak{a})|)^T \cdot N(\mathfrak{a})^{-s}$.

The study of these series is the heart of the problem.

In principle, the same arguments apply to the product $g(\mathfrak{a})\chi(\mathfrak{a})$, where χ is a Hecke character. The conclusion becomes the following. Consider only squarefree ideals $\mathfrak{a}$ = (a), with a ≡ 1 (3), for which the argument of a lies in an interval I of the unit circle. The argument of $g(\mathfrak{a})^3 = -a^2\overline{a}$ then lies in the opposite interval (-1)I, and that of $g(\mathfrak{a})$ lies in {z | $z^3 \in$ (-1)I}, that is, in the union of three disjoint intervals I_i (i = 1,2,3). The argument of $g(\mathfrak{a})$ occurs one third of the time in each of them : for $N\mathfrak{a} < X$, the number of occurrences in any one of them is $1/3(the$ number of $\mathfrak{a})$ + o(X). The same conclusion holds after imposing an additional congruence condition on a. This result may be interpreted as saying that there is no “simple” rule, even a statistical one, for determining the argument of the Gauss sum.

In [5], Heath-Brown and Patterson prove analogous results for the cubic Gauss sum $g(\mathfrak{p})$ ($\mathfrak{p}$ prime); this contradicts a suggestion of Kummer that Hasse imprudently raised to the rank of a conjecture. The Dirichlet series (0.6), with T = 1, again plays an essential role. After shifting s, it becomes

(0.7) $\sum g(\mathfrak{a})\, N(\mathfrak{a})^{-s}$;

this is the series we shall discuss.

The Dirichlet series (0.7) is not multiplicative. We now give an adelic expression for the twisted multiplicativity (0.4). For notation, see 0.0 and section 2. Let A^f be the ring of finite adeles of F, and let (,) be the cubic Hilbert symbol on $(A^f)^*$. Define a central extension $\widetilde{(A^f)}^*$ of $(A^f)^*$ by μ, equipped with a section a → [a] = (a,1), by setting $(a,\zeta')(b,\zeta'') = (ab,(a,b)\zeta'\zeta'')$. Then

(0.8) [a] [b] = [ab] · (a,b).

This is the extension defined by the 2-cocycle (a,b). Reciprocity for Hilbert symbols ensures that $[\alpha][\beta] = [\alpha\beta]$ for $\alpha, \beta \in F^*$: [ ] splits the extension over F^*. For each finite place ν of F, define in the same way an extension $\widetilde{F_\nu^*}$ of $F_\nu{}^*$ by μ ; if $\nu \nmid 3$, then [a][b] = [ab] for a,b $\in O_\nu{}^*$; if $\nu|3$, one must assume a,b $\equiv \pm 1$ (3). The restricted product of the $\widetilde{F_\nu^*}$, relative to $[O_\nu{}^*]$, is an extension of $(A^f)^*$ by $\mu^{(\Sigma)}$ ($\sum$ is the set of finite places). Pushing out by the "product" map $\mu^{(\Sigma)} \to \mu$ gives the extension $\widetilde{(A^f)}^*$.

For a character ε of μ, a function f on $\widetilde{F_\nu^*}$ (respectively $\widetilde{(A^f)}^*$) will be called an ε-function if $f(\zeta x) = \varepsilon(\zeta) f(x)$ for $\zeta \in \mu$. An ε-function is uniquely determined by its restriction to $[F_\nu{}^*]$ (respectively $[(A^f)^*]$), and it will often be identified with that restriction : its support will be described as a subset of $F_\nu{}^*$ (respectively $(A^f)^*$)....

For each ν, let f_ν be an ε-function on $\widetilde{F_\nu^*}$, right invariant under an open subgroup of $\widetilde{F_\nu^*}$. Suppose that, for almost every ν, f_ν is right invariant under $[O_\nu{}^*]$ and equals 1 on $[O_\nu{}^*]$. Under these hypotheses, the product of the $f_\nu - a$ function on the restricted product of the $\widetilde{F_\nu^*} - comes$ from an ε-function f on $\widetilde{(A^f)}^*$, right invariant under an open subgroup. An ε-function obtained in this way will be called multiplicative, and we write f = $\prod f_\nu$.

(0.9) Example : Take ε to be the identity extension of μ. For $\nu \nmid 3$, let f_ν be the ε-function on $\widetilde{F_\nu^*}$ with support {x | $v(x) = 0$ or 1}, equal to 1 on $[O_\nu{}^*]$, and given on uniformizers by

$$f_\nu([a]) = \sum_{x \in (O_\nu/(a))^*} (x,a)\, e(a^{-1}x).$$

If a is a uniformizer and $u \in O_\nu{}^*$, then $f_\nu([au]) = (u,a) f_\nu([a])$; hence f_ν is right invariant under $[O_\nu{}^*]$. For $\nu|3$, put $f_\nu([a]) = 1$ if $a \equiv 1$ (3), and $f_\nu([a]) = 0$ otherwise. Let $f = \prod f_\nu$. The Dirichlet series (0.7) is

$$\sum_{\lambda \in F^*} f([\lambda])\, N\lambda^{-s}.$$

The analytic properties of the Dirichlet series (0.7) are deduced from relations with certain "automorphic" functions on GL$(2, C)$. Identify the homogeneous space PGL$(2, C)/U(2)$ with upper half-space $H = \{(z, v) \mid z \in C, v \in R, v > 0\}$ by the Iwasawa decomposition

$$(z,v) \mapsto \begin{pmatrix} 1 & z \\ 0 & 1 \end{pmatrix}\begin{pmatrix} v & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} v & z \\ 0 & 1 \end{pmatrix}.$$

In this model, the upper unipotent group acts by $(z, v) \to (z + t, v)$, and

$$\begin{pmatrix} a & 0 \\ 0 & 1 \end{pmatrix}(z,v) = (az, |a|v),\ a \in C^*.$$

Let $\Gamma(3)$ be the subgroup of GL$(2, O)$ consisting of matrices congruent to 1 modulo 3. Kubota discovered that one obtains a character of this group by setting, for

γ = (a b),
(c d)

$\chi(\gamma) = (c/a)_3$ if c $\neq$ 0, and $\chi(\gamma) = 1$ otherwise.

Put $\Gamma_2 = \Gamma(3) \cdot \mathrm{GL}(2, Z)$. Patterson verifies that χ extends to a character of Γ_2 that is trivial on $\mathrm{GL}(2, Z)$; this extension is again denoted χ. The kernel of χ is visibly not a congruence subgroup. Nevertheless,

(0.10) For every $\delta \in \mathrm{GL}(2, F)$, the characters $\chi(\gamma)$ and $\chi(\delta^{-1}\gamma\delta)$ coincide on a congruence subgroup of $\Gamma_2 \cap \delta\Gamma_2\delta^{-1}$.

Again denote by ε the identity embedding of μ, and introduce the multiplicative ε^2-function $\tau = \prod \tau_\nu$ characterized by the following properties (cf. [18] I, Th. 8.1, p. 152). (a) For $\nu \nmid 3$, τ_ν is supported in O_ν and is right invariant under $[O_\nu{}^*]$. (b) For $\nu|3$, τ_ν is supported in {x | $\nu(x) \geq \{\} - 3$} and is right invariant under [{x | $x \equiv \pm 1$ (3)}]. (c) For d in O_ν, and x lying respectively over O_ν when $\nu \nmid 3$ and over {x | $\nu(x) \geq \{\} - 3$} when $\nu|3$, one has $\tau_\nu(x[d^3]) = \tau(x)\|d\|_\nu$. (d) If $\nu \nmid 3$, τ_ν equals 1 on $[O_\nu{}^*]$, equals $\|a\|_\nu f_\nu(a)^-$ on uniformizers (f_ν as in 0.10), and equals 0 on elements of valuation 2. (e) If $\nu|3$ and $x = \pm\omega^j\lambda^k$, where $\omega = \exp(2\pi i/3)$ and $\lambda = \sqrt{-3}$, then

$\tau_\nu(x) = 1$ if x = $\pm\lambda^{-3}$;
$\tau_\nu(x) = 1/3$ exp(-j $2\pi i/9$) if x = $\pm\omega^j\lambda^{-1}, j = -1$, 0, or 1;
$\tau_\nu(x) = 0$ if (j,k) is not congruent modulo 3 to
(0,0), (0,2), (1,2), or (2,2).

For $w = (z, v) \in H$, put

$$K(w) = v\, K_{1/3}(4\pi v)e(z),$$

where $K_{1/3}$ is the "usual Bessel function of the second kind" ([18] I, pp. 129, 134).

One of the principal results of [18] is that the series

(0.11) $\theta(w) = \sigma v^{2/3} + \sum$ $\tau([\mu])$ K((μ 0) w),
$\mu \in F^*$ (0 1)

$$\sigma = 2^{-1}3^{-3/2},$$

satisfies $\theta(\gamma w) = \chi(\gamma)\theta(w)$ for $\gamma \in \Gamma_2$ (and in particular $\theta(0, v) = \theta(0, v^{-1})$), and is the residue (in s) of an Eisenstein series $E_s - defined$ by analytic continuation in a parameter s—at a value of s where a simple pole occurs. Up to a factor, the Eisenstein series is obtained from $\varphi_s(w) = v^s$ by summing $\overline{\chi(\gamma)}\varphi_s(\gamma w)$ over $\gamma \in \Gamma_2/\Gamma_2 \cap B^+$ (notation : see $0.0.1$). The pole is at $s = 4/3$.

Up to elementary factors, Γ and ζ, the Dirichlet series (0.7) is obtained from $\theta(w)$ by Mellin transformation. This gives its analytic behavior and a functional equation.

Patterson calls θ a "cubic θ-series," by analogy with the classical case of holomorphic modular forms of weight $1/2$, where the series $\theta(z) = \sum q^{n^2}$ $(q = e^{2\pi iz})$ has an analogous description as a residue of Eisenstein series.

Some of these results can be predicted by the representation theory of covers of the groups $\mathrm{GL}(2, F_v)$. Here is the method. As above, put $\varphi_s(w) = v^s$ and, for a congruence subgroup Γ, consider the Eisenstein series

$$E_{s,\Gamma}(w) = \sum_{\gamma \in \Gamma/\Gamma \cap B^+} \varphi_s(\gamma w).$$

It converges for Re(s) large and admits analytic continuation as a meromorphic function of s. The study of the "constant terms" shows that, for $Re(s) \geq 4/3$, it is meromorphic in s, with at worst a simple pole at $s = 4/3$. Let E_Γ be the residue at $s = 4/3$, and let $\mathscr{E}$ be the space of linear combinations of the functions $E(\gamma w)$, for congruence subgroups Γ and $\gamma \in \mathrm{GL}(2, F)$.

The group $\mathrm{GL}(2, F)$ acts on $\mathscr{E}$ by $(\gamma * f)(w) = f(\gamma^{-1}w)$. For every $f \in \mathscr{E}$ there is a congruence subgroup $\Gamma \subset \Gamma(3)$ such that $\gamma * f = \chi(\gamma)f$ for $\gamma \in \Gamma$. This follows from (0.10). In particular, $\gamma * f = f$ for $\gamma \in \Gamma \cap \mathrm{Ker}(\chi)$, which allows the action of $\mathrm{GL}(2, F)$ on $\mathscr{E}$ to be extended continuously to an action of the completion $\mathrm{GL}(2, F)$^ of $\mathrm{GL}(2, F)$ for the topology defined by the subgroups $\Gamma \cap \mathrm{Ker}(\chi)$, with $\Gamma \subset \Gamma(3)$ a congruence subgroup. The study of the "constant terms," together with a "local" study, shows that the representation $\mathscr{E}$ is irreducible and determines its isomorphism class.

For $f \in \mathscr{E}$ satisfying $f(z, v) = f(z + \lambda, v)$ for λ in a lattice $L \subset O$, define the function $W^*(f)$ on H by

$W^*(f)(z, v) = 1/vol(C/L) \int_{C/L} f(z + z_1, v)e(-z_1)dz_1.$

This function is independent of the choice of L. Since φ_s, and hence $E_{s,\Gamma}$, are eigenfunctions of the Laplacian, $W^*(f)$ is K(w) times a constant W(f). Fourier inversion on C/L, together with a study of the constant terms, gives an expansion

$$f = \sigma(f)v^{2/3} + \sum_{\mu \in F^*} W\left(\begin{pmatrix} \mu & 0 \\ 0 & 1 \end{pmatrix} * f \right) K\left(\begin{pmatrix} \mu & 0 \\ 0 & 1 \end{pmatrix} w\right),$$

for a suitable constant $\sigma(f)$.

The functional W is a linear form on $\mathscr{E}$. It satisfies

$$W\left(\begin{pmatrix} 1 & \nu \\ 0 & 1 \end{pmatrix} * f \right) = e(\nu)W(f).$$

for $\nu \in$ F. This, essentially (up to a nuisance due to the place dividing 3), determines it up to a scalar - hence a determination of the nonconstant terms in the Fourier expansions of the f in $\mathscr{E}$.

These methods give the following result. Let T be the set of linear combinations of multiplicative ε^2-functions $\tau' = \prod \tau'_\nu$ on $(\widetilde{A^f})^*$, where a) for almost every ν, one has $\tau'_\nu = \tau_\nu$; b) for every ν, τ'_ν has compact support, or is a right translate of τ_ν. For each $\tau' \in$ T there is then a unique constant $\sigma(\tau')$ such that the function

$$\theta(\tau'; w) = \sigma(\tau')\, v^{2/3} + \sum_{\mu \in F^*} \tau'([\mu])\, \mathrm{K}\left(\begin{pmatrix} \mu & 0 \\ 0 & 1 \end{pmatrix} \mathrm{w} \right)$$

satisfies $\theta(\tau'; \gamma w) = \chi(\gamma)\, \theta(\tau'; w)$, for γ in a suitable congruence subgroup. The space $\mathscr{E}$ is the set of the $\theta(\tau'; w)$.

The group $\mathrm{GL}(2, F)^\wedge$ is not a restricted product of local groups. It may, however, be compared with such a product, thereby reducing the proof of the preceding results to the study of local groups and of the “principal series” of their representations. The main part of the exposé is devoted to this study.

The method followed by Patterson [18] is different. He begins by proving functional equations for Dirichlet series such as (0.7), and from them deduces the automorphic character of θ by a variant ([18] 7.1) of Weil’s method [23]. He deduces the required functional equations from the functional equation (for s $\to$ 2 - s) of Eisenstein series ([11],[12]), by examining what it implies for the Fourier coefficients : each of them, as a function of s, is a Dirichlet series reminiscent of (0.7). He hopes that this method will also provide the required analytic properties of Dirichlet series analogous to (0.7), in order to generalize [5] to cases where n > 3. The method presented here, by contrast, depends crucially on n being odd and on $(n-1)/2 = 1$.

In the text we shall dwell on the local theory and only sketch how it applies to residues of Eisenstein series. We shall discuss neither the double role, indicated above, of series (0.7), for me one of the great mysteries of the theory, nor the determination of the constant term, nor the ingenious method by which Patterson [18] identifies θ, proved automorphic, with the residue of the Eisenstein series.

0.0 **Notations :**

0.0.1 We shall denote the following subgroups of GL(2) as follows :
Z : the center (scalar matrices).
T : diagonal matrices. We put diag (a,b) = (a 0).
(0 b)
B,U : Borel subgroup ; its unipotent radical. In particular :

B^+ : (* *) ; B^- : (* 0) ; U^+ : (1 *) ; U^- : (1 0).
(0 *) (* *) (0 1) (* 1)

0.0.2 From section 2 onward, F denotes either a local field (= locally compact and non-discrete) or a global field (= a finite extension of Q or of a field $F_p(t)$). We choose an integer n, not divisible by the characteristic of F, such that F contains a primitive n-th root of unity. We write $\mu_n(F)$, or simply μ, for the group of n-th roots of unity in F.

For global F and a place v of F, the inclusion of F in its completion F_v induces an isomorphism from $\mu_n(F)$ to $\mu_n(F_v)$, by means of which we identify these two groups.

0.0.3 For a global field F, we write A for its ring of adeles, F_∞ for the product of the F_v over the real or complex places v, and A^f for the ring of finite adeles :

$$\mathrm{A} = F_\infty \times A^f.$$

If F is a number field, O denotes its ring of integers. If F is a nonarchimedean local field, O denotes its valuation ring. Variant : notation F_v, O_v.

0.0.4 For a local (respectively global) field F, from section 3 onward we shall have to consider a central extension of G(F) (respectively G(A)) by μ. A ~ denotes the inverse image of a subgroup in this extension. For example : $T(F)^\sim$, $T(A)^\sim$, $G(A^f)^\sim$, ...

0.0.5 From section 4 onward, fix a faithful character $\varepsilon : \mu \hookrightarrow C^*$. A representation of a central extension of a group by μ will be called an ε-representation if each $\zeta \in \mu$ acts by multiplication by $\varepsilon(\zeta)$. A function on the extension will be called an ε-function if $f(\zeta x) = \varepsilon(\zeta) f(x)$ for $\zeta \in \mu$.

0.0.6 A representation π of a totally disconnected locally compact group is called algebraic if the stabilizer of every vector is an open subgroup, and admissible if, in addition, the subspace fixed by an open subgroup is finite-dimensional. For the extension of this terminology to other groups, such as $\mathrm{GL}(2,A)^\sim$, I refer to [6].

0.0.7 From section 5 onward, it will sometimes be useful to write the multiplication of characters additively, and to write s for the character $\|x\|^s$; for example, to write $(\alpha_2 - \alpha_1 - 1)(\lambda)$ for $\alpha_2(\lambda)\ \alpha_1(\lambda)^{-1}\ \|\lambda\|^{-1}$. The real part of a character σ is the real number s such that $\sigma(x)\|x\|^{-s}$ is unitary.

1. Central extensions of $\mathrm{SL}(2)$

Let F be a field. Recall the

Definition 1.1 : (i) A symbol is a map s from $F^* \times F^*$ to an abelian group (written multiplicatively), which is bimultiplicative and satisfies s(a,b) = 1 when a + b = 1.

(ii) $K_2(F)$ denotes the quotient of $F^* \otimes F^*$ by the subgroup generated by the a ⊗ b, for a,b ∈ F^* with a + b = 1, and {x,y} denotes the image in $K_2(F)$ of x ⊗ y; it is the universal symbol.

Example 1.2 Let n be an integer ≥ 1, prime to the characteristic exponent of F, and suppose that F contains the n-th roots of unity. Let F̄ be a separable closure of F. The long exact $\mathrm{Gal}(\overline{F}/F)$-cohomology sequence deduced from the Kummer exact sequence $1 \to \mu_n \to \overline{F}^* \to \overline{F}^* \to 1$ gives isomorphisms $F^*/F^{*n} \simeq H^1(\mu_n)$ and $H^2(\mu_n) \simeq \mathrm{Br}(F)_n$, and the cup product

$$H^1(\mu_n) \otimes H^1(\mu_n) \to H^2(\mu_n \otimes \mu_n) = H^2(\mu_n) \otimes \mu_n$$

gives a bimultiplicative map from $F^* \times F^*$ to $\mathrm{Br}(F)_n \otimes \mu_n$. It is a symbol ([21] XIV 2).

1.3 Let G be an algebraic group over F, isomorphic to $\mathrm{SL}(2)$. There is then a canonical central extension $G(F)^\wedge$ of G(F) by $K_2(F)$. For every symbol s : $F^* \times F^* \to$ A, it yields a central extension $G(F)^s$ of G(F) by A.

I did not take G = $\mathrm{SL}(2)$ from the outset so that I could say that the adjoint group $\mathrm{PGL}(2,F)$ acts on $\mathrm{SL}(2,F)^s$ by transport of structure. This action lifts the action by "inner" automorphisms of $\mathrm{PGL}(2,F)$ on $\mathrm{SL}(2,F)$ (rather than "inner," one should say : induced by an inner automorphism of $\mathrm{GL}(2,F)$). It is trivial on A.

Remark 1.4 (Not used below). Exclude the cases in which F has 2 or 3 elements. Then $\mathrm{SL}(2,F)$ is perfect, and according to Moore [17] and Matsumoto [14], the universal central extension of $\mathrm{SL}(2,F)$ is a central extension by the symplectic analogue ${}_{-1}L_2(F)$ of $K_2(F)$. The group $\mathrm{PGL}(2,F)$ acts on ${}_{-1}L_2(F)$ (through its quotient by $\mathrm{PSL}(2,F)$), and $K_2(F)$ is the group of coinvariants (the largest quotient inheriting a trivial action). According to Karoubi [24] III p. 78, if F is not of characteristic 2, the group ${}_{-1}L_2' := \mathrm{Ker}({}_{-1}L_2(F) \to K_2(F))$ fits into an exact sequence

$F^* - - -1, x - - >$ (2-torsion of $K_2(F)$) $\to {}_{-1}L_2' \to$ W $- - r - - > K_2(F)/2K_2(F) \to 0$,

where W is the group of virtual quadratic forms of dimension 0 and discriminant 1. I imagine that r coincides with the Stiefel-Whitney class defined by Milnor [15] ; according to Milnor, $\mathrm{Ker}(r)$ would then be the cube of the augmentation ideal (dim = 0) in the ring of virtual quadratic forms.

1.5 The group $\mathrm{GL}(2, F)$ is the semidirect product of the subgroup of diagonal matrices $\mathrm{diag}(a, 1)$ by the normal subgroup $\mathrm{SL}(2, F)$. Likewise take the semi-direct product of F^* by $\mathrm{SL}(2, F)^s$, with a ∈ F^* acting as the diagonal matrix $\mathrm{diag}(a, 1)$ in $\mathrm{PGL}(2, F)$. This is a central extension $\mathrm{GL}(2, F)^s$ of $\mathrm{GL}(2, F)$ by A. For s the universal symbol, it will be denoted $\mathrm{GL}(2, F)^\wedge$.

After the preliminaries in 1.6, we shall give in 1.7-1.9 some properties of these central extensions. They suffice to characterize them.

1.6 Preliminaries : (a) If $H^\sim$ is a central extension of a group H, and a,b ∈ H, the commutator (ã,b̃) of lifts ã and b̃ of a and b depends only on a and b. We denote it simply by (a,b). Likewise, the inner automorphism $int_{\tilde{a}}$ of $H^\sim$ depends only on a. We denote it simply by int_a.

(b) If U is a subgroup of H, normalized by B ⊂ H and generated by the commutators (b,u) (b ∈ B, u ∈ U), there is at most one lift u → ũ of U into $H^\sim$, normalized by B. If the homomorphism u → ũ commutes with int_b (b ∈ B), the lift of a commutator (b,u) is necessarily, as in (a) above,

$$(b, u)^\sim = int_b(u)^\sim\, \tilde{u}^{-1} = int_b(\tilde{u}) \cdot \tilde{u}^{-1} = (\tilde{b},\tilde{u}).$$

1.7 Excluding the cases in which F has 2 or 3 elements, 1.6(b) applies to a Borel subgroup B of $\mathrm{SL}(2, F)$ and its unipotent radical U, i.e. to the stabilizer B of a (homogeneous) line D ⊂ F^2 and to its pointwise stabilizer U. It is easy to show that, for every central extension of $\mathrm{SL}(2, F)$, U has a lift normalized by B (unique by 1.3). We write it u → ũ. One has ẽ = e, and if u ≠ e fixes a line, that line is uniquely determined. Thus the map u → ũ is well defined on the union of the conjugates of U (the unipotent elements). It is invariant under conjugation. In the case of the extension 1.3, it commutes with the action of $\mathrm{PGL}(2, F)$.

1.8 The group $\mathrm{GL}(2, F)^s$ was defined as a semidirect product $F^* \times \mathrm{SL}(2, F)^s$. We denote by d_1 the corresponding inclusion of F^*, and by d_2 its conjugate by

w = (0 1) (cf. 1.9).
(-1 0)

For a,b ∈ F^*, $d_1(a)$ and $d_2(b)$ are respectively lifts of $\mathrm{diag}(a, 1)$ and $\mathrm{diag}(1, b)$ in $\mathrm{GL}(2, F)$. These matrices commute. Hence the commutator of $d_1(a)$ and $d_2(b)$ lies in A. One has

$$(d_1(a), d_2(b)) = \mathrm{s(a,b)},$$

and this formula uniquely determines the extension of $\mathrm{SL}(2, F)$ under consideration.

Here is a more complete formula sheet.

1.9 **Formula sheet : Put**

$e^+(a)$:= the lift of 1.7 of (1 a) $e^-(a)$:= the lift of 1.7 of (1 0)
(0 1) (a 1)

w(a) := $e^+(a)e^-(-1/a)e^+(a)$, a lift of $\begin{pmatrix} 0 & a \\ a^{-1} & 0 \end{pmatrix}$

$d_2(a)$:= w(1) $d_1(a)$ $w(1)^{-1}$
h(a) := $d_1(a)$ $d_2(a^{-1})$
For every a ∈ F^*, the images in SL$(2, F)$ of $e^+ = e^+(a)$, $e^- = e^-(-1/a)$, and w = w(a) belong to the Tits trijection. In SL$(2, F)$^ one has
(1.9.1) $e^\pm = int_w(e^\mp)$
(1.9.2) w = e^+ e^- e^+ = e^- e^+ e^-
(expressing $int_w(w)$ = w). Each of the elements w, e^+, e^- determines the other two. It follows that w(a) = $d_1(a)w(1)d_1(a)^{-1}$, and
(1.9.3) w(a) $w(1)^{-1}$ = $d_1(a)$ $d_2(a^{-1})$ = : h(a).
In terms of h, formula
(1.9.4) $(d_1(a), d_2(b))$ = s(a,b)
may be rewritten as
(1.9.5) h(ab) = s(a,b) h(a) h(b).
Example 1.10 Take F = R, A = ±1, and s(a,b) = 1 if a or b > 0, and -1 otherwise. The extension SL$(2, R)^s$ is then the nontrivial double cover SL$(2, R)$~ of SL$(2, R)$. It has the following description :
(a) Let $(R^2$ - 0)~ be a nontrivial double cover of R^2 - {0}. Since $\pi_1(R^2$ - {0}) = Z, it is unique up to isomorphism.
(b) The group SL$(2, R)$~ is the group of elements of SL$(2, R)$, equipped with a lift to $(R^2$ - 0)~ of their action on R^2.
The same description holds for the cover GL$(2, R)^s$ of GL$(2, R)$.
1.11 An analogous description applies over an arbitrary field and, without being indispensable, is convenient for computation. It requires part of the general theory of the Zariski sheaves $\underline{a}K_q$ on a scheme S associated with the presheaves U → $K_q(\Gamma(U, O))$. Specifically, we shall use special cases of the following isomorphism, deduced from Quillen's resolution of $\underline{a}K_q$ (the Gersten conjecture) : for T smooth of pure codimension d in a scheme S smooth over F, one has

$$(1.11.1)\ \underline{H}^i_T(S, \underline{a}K_q) = \begin{cases} \underline{a}K_{q-d} & \text{for } i = d \\ 0 & \text{for } i \neq d \text{ (and if } d > q). \end{cases}$$

Special case : (a) for the affine plane A^2 = Spec F[X,Y],
(1.11.2) $H^0(A^2$ - {0}, $\underline{a}K_2) = K_2(F)$

(1.11.3) $H^1(A^2 - \{0\}, \underline{a}K_2) = \mathrm{Z}\ (= K_0(F))$.

(b) For T of codimension 1 in S, the case i = 1 of (1.11.1) gives a residue morphism Res : $H^0(S - T, \underline{a}K_2) \to H^0(T, \underline{a}K_1) = H^0(T, O_T^*)$. Suppose T is connected, and let v denote the corresponding valuation; then

$$\mathrm{Res}\{f,g\} = (-1)^{v(f)v(g)} f^{v(g)}\ g^{-v(f)} \text{ restricted to T.}$$

The residue morphism vanishes on the image of $H^0(S, \underline{a}K_2)$. If P is a torsor under $\underline{a}K_2$ on S and e is a section of P on S - T, this also makes it possible to define Res e $\in H^0(T, O_T^*)$, which vanishes if and only if e extends across T.

Now let P be a torsor under $\underline{a}K_2$ on A^2 - $\{0\}$. Since the isomorphism (1.11.3) is canonical, P is isomorphic to its transform by any element of $\mathrm{GL}(2, F)$, and, by (1.11.2), the group of elements of $\mathrm{GL}(2, F)$, equipped with a lift to the torsor P of their action on A^2 - $\{0\}$, is a central extension of $\mathrm{GL}(2, F)$ by $K_2(F)$. We shall verify that, when the class of P is a generator of $H^1(A^2 - 0, \underline{a}K_2)$ = Z, this is the extension $\mathrm{GL}(2, F)$^.

1.12 With respect to the cover of A^2 - $\{0\}$ by the open sets Y $\neq$ 0 and X $\neq$ 0, the generator of H^1 is represented by the Čech cocycle {X,Y}. Take P to be the torsor equipped with a section e_X on X $\neq$ 0 and a section e_Y on Y $\neq$ 0, with

$$e_X - e_Y = \{X,Y\}.$$

Note that e_Y - e_X = {Y,X}, so that this description of P is symmetric in the coordinates X and Y. One has

$$\mathrm{Res}\ e_X = \mathrm{Res}\{X,Y\} = Y^{-1} \text{ (on X = 0),}$$

and symmetrically. To identify the central extension of $\mathrm{GL}(2, F)$ defined by P with the extension $\mathrm{GL}(2, F)$^, set

$$d_1(a) = \text{the lift to P of } \mathrm{diag}(a, 1) \text{ that is the identity}$$

at the fixed point (0,1); hence, since w sends (0,1) to (1,0),

$$d_2(a) = \text{the analogous lift for } \mathrm{diag}(1, a) \text{ and (1,0).}$$

To verify (1.9.4), note that $d_1(a)$ transforms e_Y into e_Y + {a,Y} (whose residue is the transform aX^{-1} of the residue X^{-1} of e_Y), whereas $d_2(a)$ transforms e_Y into itself.

1.13 Let $A^{2\vee}$ be the affine scheme Spec F[a,b], corresponding to the dual of the natural representation of $\mathrm{GL}(2, F)$. To $\ell \in A^{2\vee}$ - $\{0\}$, associate the group of trivializations of P on ℓ = 1. It is a principal homogeneous space under $K_2(F)$, and, by sheafifying this construction, one obtains a torsor $P^\vee$ under $\underline{a}K_2$ on $A^{2\vee}$ - $\{0\}$. By construction, P and $P^\vee$ become isomorphic on the locus $\ell(x) = 1$

of $A^{2\vee} \times A^2$, and $\mathrm{GL}(2, F)$ acts on P and $P^\vee$ while respecting this isomorphism.
Here is a more complete formula sheet.

1.14 **Formula sheet :**

P : a torsor on A^2 - $\{0\}$, where A^2 is the plane with coordinates X,Y. $P^\vee$: a torsor on $A^{2\vee}$ - $\{0\}$, where $A^{2\vee}$ is the dual plane, with coordinates a,b. e_X, e_Y : sections of P on X $\neq$ 0, Y $\neq$ 0. e_a, e_b : sections of $P^\vee$ on a $\neq$ 0, b $\neq$ 0.

The defining formulas are (a) e_X - e_Y = {X,Y} ; (b) P and $P^\vee$ become isomorphic on the locus aX + bY = 1 in $A^2 \times A^{2\vee}$. Under this isomorphism, e_a at (a,b) coincides with e_X at $(1/a, 0)$, and e_b at (a,b) coincides with e_Y at $(0, 1/b)$.

One checks that the resulting formula sheet is symmetric in P and $P^\vee$ and in the coordinates : it is preserved by the substitutions (X,Y ;a,b) $\mapsto$ (a,b ;X,Y) and (Y,X ;b,a). One has

(1.14.1) e_X - e_Y = {X,Y} and e_a - e_b = {a,b} ;

(1.14.2) e_a - e_Y = -{a,Y} on the locus aX + bY = 1, and likewise

$$e_b - e_X = -\{b,X\} ;$$

(1.14.3) $d_1(x)_*$ and $e^-(y)_*$ fix e_X and e_a,
and likewise

$$d_2(x)_* \text{ and } e^+(y)_* \text{ fix } e_Y \text{ and } e_b.$$

2. Review : the Hilbert symbol

The notation of 0.0.2 is in force.

2.1 Let F be a local field. If F is not complex, then $\mathrm{Br}(F)_n = Z/nZ$, and the symbol of 1.2 is the Hilbert symbol

$$\{\ ,\ \} : F^* \times F^* \to \ \mu.$$

For complex F, the Hilbert symbol is defined to be trivial. As n varies, these symbols, denoted $\{\ ,\ \}_n$, satisfy $a, b_n = a, b_{nm}^m$.

Recall 2.2 : The pairing induced by the Hilbert symbol,

$$\{\ ,\ \} : F^*/F^{*n} \times F^*/F^{*n} \to \ \mu,$$

is nondegenerate : it identifies each factor F^*/F^{*n} with the Pontryagin dual $\mathrm{Hom}(F^*/F^{*n},\ \mu_n(F))$ of the other.

In the definition of $\{\ ,\ \}_n$ we adopt Serre's signs [21], i.e. those for which (2.4.1) below is true.

Example 2.3 : F = R, n = 2 : this is the symbol of 1.10.

Example 2.4 : The tame case is that in which F is non-Archimedean and n is prime to its residual characteristic. Place ourselves in this case, and let O be the valuation ring of F, v its valuation, k = O/m the residue field, q = $|k|$, and r the reduction map modulo m. One has n | q-1 ; r induces an isomorphism $\mu_n(F) \simeq \ \mu_n(k)$, and

(2.4.1) $ra, b_n = r((-1)^{v(a)v(b)}\ a^{v(b)}\ b^{-v(a)})^{(q-1)/n}$.

2.5 Let F be a global field. Denote by $\{\ ,\ \}_\nu\{\}$ the Hilbert symbol for the completion F_ν of F at a place ν. It takes values in $\mu_n(F_\nu)$, identified with μ by (0.0.2). For two idèles x and y of F, one has $x_\nu, y_{\nu_\nu} = 1$ at almost every place ν (cf. 2.4). This makes it possible to define {x,y} as the product of the x_ν, y_{ν_ν} over all places.

Recall 2.6 : The induced pairing

$$\{\ ,\ \} : A^*/A^{*n} \times A^*/A^{*n} \to \ \mu$$

is nondegenerate. It identifies the locally compact group A^*/A^{*n} with its own Pontryagin dual $\mathrm{Hom}(A^*/A^{*n},\ \mu)$.

This follows from 2.2 and from the fact that in the tame case 2.4, hence for almost every ν, the image of $O_\nu{}^*$ in $F_\nu{}^*/F_\nu{}^{*n}$ is maximal totally isotropic for $\{\ ,\ \}_\nu\{\}$.

The reciprocity law for Hilbert symbols says that if the idèles x and y are principal (x,y ∈ F^*), then {x,y} = 1. More precisely :

Recall 2.7 : (i) $F^{*n} = F^* \cap A^{*n}$. (ii) The image F^*/F^{*n} of F^* in A^*/A^{*n} is maximal totally isotropic. It is a discrete subgroup with compact quotient, and { , } identifies the Pontryagin dual of the discrete group F^*/F^{*n} with the quotient $A^*/A^{*n} \cdot F^*$.

(ii) The Image F^*/F^{*n} of F^* in A^*/A^* is maximal totally isotropic. It is a discrete subgroup with compact quotient, and { , } identifies the Pontryagin dual of the discrete group F^*/F^{*n} with the quotient $A^*/A^{*n} \cdot F^*$.

Example 2.8 Take for F the field $Q(\surd -3)$ of cube roots of unity, and let n = 3. Let K be the compact open subgroup of $A^*/A^{*3} = A^{f*}/A^{f*3}$ that is the image of the product of the $O_\nu{}^* \subset F_\nu{}^*$ for $\nu \nmid 3$, and of $\{x \in F_\nu{}^* \mid x \equiv 1(3)\}$ for $\nu \mid 3$. The group A^*/A^{*3} is the product of the discrete group F^*/F^{*3} and the compact group K; each of these two subgroups is maximal isotropic, and { , } identifies either one with the Pontryagin dual of the other.

3. The metaplectic group

3.1 Let F be a local field. A ~ will denote the central extensions of $\mathrm{SL}(2,F)$ and $\mathrm{GL}(2,F)$ by μ deduced from the Hilbert symbol.
3.2 In the tame case (2.4), the central extension $\mathrm{SL}(2,F)^{\sim}$ of $\mathrm{SL}(2,F)$ is trivial over the maximal compact subgroups. The trivialization is moreover unique : if the residue field does not have 2 or 3 elements, $\mathrm{SL}(2,O)$ is perfect and, in all cases, its largest abelian quotient has order prime to n. Over the matrices (1 a) and (1 0) (a $\in$ O), it coincides with the trivialization of 1.7. (0 1) (a 1)

Likewise, the central extension $\mathrm{GL}(2,F)^{\sim}$ of $\mathrm{GL}(2,F)$ is trivialized over $\mathrm{GL}(2,O)$, by the unique trivialization extending $\mathrm{diag}(a,1) \mapsto d_1(a)$ (a $\in$ O^*).
3.3 Now take F global. At almost every place ν one is in the tame case, and 3.2 makes it possible to define the restricted product $\mathrm{SL}(2,A)^{\sim}$ of the $\mathrm{SL}(2,F_\nu)^{\sim}$ relative to the compact subgroups lifting $\mathrm{SL}(2,O_\nu)$. This is an extension of $\mathrm{SL}(2,A)$ by the direct sum of the $\mu_n(F_\nu)$. Pushing out by the product map $\prod : \oplus\ \mu_n(F_\nu) \to\ \mu_n(F) =: \mu$, one obtains a central extension $\mathrm{SL}(2,A)^{\sim}$ of $\mathrm{SL}(2,A)$ by μ, and the reciprocity law for Hilbert symbols ensures that this extension splits over $\mathrm{SL}(2,F)$. Since $\mathrm{SL}(2,F)$ is perfect, the splitting is unique. On the unipotent radical of a Borel subgroup it is induced by the splitting obtained from the local splittings (1.7) :

$$\begin{array}{c}
1 \to\ \mu \to\ \mathrm{SL}(2,F_\nu)^{\sim} \to\ \mathrm{SL}(2,F_\nu) \to\ 1 \\
\Vert \downarrow\ \downarrow \\
(3.3.1)\ 1 \to\ \mu \to\ \mathrm{SL}(2,A)^{\sim} \to\ \mathrm{SL}(2,A) \to\ 1 \\
\nwarrow\ [\,]\uparrow \\
\mathrm{SL}(2,F)
\end{array}$$

3.4 Take n = 3, F = $Q(\sqrt[3]{1}) \subset$ C, and ε = Id. One has $\mathrm{SL}(2,A)^{\sim} = \mathrm{SL}(2,C) \times \mathrm{SL}(2,A^f)^{\sim}$, whence, with the notation of the Introduction,

$$\mathrm{SL}(2,A)^{\sim} / \mathrm{SU}(2) \times \{1\} \simeq\ \mathrm{H} \times \mathrm{SL}(2,A^f)^{\sim};$$

since $\mathrm{SL}(2,C)/SU(2) \simeq\ \mathrm{PGL}(2,C)/U(2)$. Restriction to H × $\{1\}$ therefore associates with every function f on $\mathrm{SL}(2,A)^{\sim}$ that is right-invariant under SU(2) a function on H.

Scholium 3.5 The restriction map above gives a bijective correspondence between (a) the space of ε-functions on $\mathrm{SL}(2,A)^{\sim}$, left-invariant under $[\mathrm{SL}(2,F)]$ and right-invariant under SU(2) × (a suitable open subgroup K of $\mathrm{SL}(2,A^f)^{\sim}$),

(b) the space of functions on H which, for γ in a suitable congruence subgroup Γ of $\mathrm{SL}(2,O)$, satisfy

$$f(\gamma w) = \chi(\gamma)\ \mathrm{f(w)}.$$

The point is that $\mathrm{SL}(2, A^f)^\sim$ is identified with the completion of $\mathrm{SL}(2, F)$ for the topology whose subgroups are the kernels of χ inside congruence subgroups of $\mathrm{SL}(2, O)$. See [14].

The group $\mathrm{SL}(2, A^f)^\sim$ acts by right translations on the space in (a). From the classical point of view (b), this corresponds to, and makes more precise, the existence of Hecke operators. To express and use (0.11), it is moreover useful to pass to $\mathrm{GL}(2)$.

3.6 For a global field F, again arbitrary, define $\mathrm{GL}(2, A)^\sim$ in either of two ways : (a) by pushing out the restricted product of the $\mathrm{GL}(2, F_v)^\sim$ by $\sum : \oplus\ \mu_n(F_v) \to\ \mu$; (b) as the semidirect product of A^* by $\mathrm{SL}(2, A)^\sim$.

There is again a diagram $(3.3.1)$, with SL replaced by GL ; the splitting [ ] coincides with d_1 on $\mathrm{diag}(a, 1)$.

3.7. Suppose again that F is a local field. From the Hilbert symbol one deduces a morphism of Zariski sheaves on $A^2 - 0$, from $\underline{a}K_2$ to the sheaf of locally constant functions on $F^2 - 0$, for the usual topology, with values in μ. This makes it possible to deduce from P (1.11) a torsor P_n under μ on $F^2 - 0$, equipped with its usual topology. For each homogeneous line D there is a set of distinguished trivializations of P_n on F^2-D : those that are images of a trivialization of P on A^2-D. If d = 0 is an equation of D, any two differ by a function $\zeta \cdot a, d(x)$ with values in μ.

The description 1.11 of $\mathrm{GL}(2, F)^\wedge$ gives here : $\mathrm{GL}(2, F)^\sim$ is the group of elements of $\mathrm{GL}(2, F)$ equipped with a lift to P_n of their action on F^2, this lift respecting the system of distinguished trivializations above.

3.8 In particular, $Z(F)^\sim$ acts on P_n. The action will be written as scalar multiplication : $z \cdot p$. For a,b $\in\ F^{*n}$, put $diag^\sim(a, b) = d_1(a)d_2(b)$; $diag^\sim$ is a homomorphism $F^{*n} \times F^{*n} \to T(F)^\sim$. For a $\in\ F^{*n}$ and p $\in\ P_n$, put $a \cdot p = diag^\sim(a, a)p$. This formula lifts to P_n the homothety of ratio a on F^2.

3.9 As in 1.13, there is a torsor $P_n^\vee$ on the dual and an isomorphism between P_n and $P_n^\vee$ on the locus $\ell(x) = 1$, all $\mathrm{GL}(2, F)^\sim$-equivariantly. Formula sheet 1.14 remains valid.

For a $\in\ F^{*n}$, write $a \cdot p$ for the action of $\mathrm{diag}(a^{-1}, a^{-1})$ on $P_n^\vee$; it lifts to $P_n^\vee$ the homothety v $\mapsto$ av on $F^{2\vee}$.

3.10 **In the rest of the exposé, the torsors P_n and $P_n^\vee$ will simply be denoted P and $P^\vee$.**

3.11 For global F, and at almost every place ν, there is a trivialization of P_ν on $(A^2 - 0)(O_\nu) = ({O_\nu}^* \times O_\nu) \cup (O_\nu \times {O_\nu}^*)$: for tame ν, e_X on ${O_\nu}^* \times O_\nu$ and e_Y on $O_\nu \times {O_\nu}^*$ coincide on ${O_\nu}^* \times {O_\nu}^*$. This makes it possible to adelize 3.7 and obtain a torsor P under μ on $(A^2 - 0)(A)$. The action of $\mathrm{GL}(2, A)$ on A^2 lifts to an action of $\mathrm{GL}(2, A)^\sim$ on this torsor. The torsor is trivialized on $(A^2 - 0)(F) = F^2 - 0$ by e_X and e_Y, and $[\mathrm{GL}(2, F)]$ respects this trivialization.

4. The group of diagonal matrices.

4.1 Heisenberg groups : Let $H^{\sim}$ be a locally compact group that is an extension of a locally compact, totally disconnected, commutative group H by its center Z. Fix a character ω of Z. Assume that ω is unitary on commutators, and that the bicharacter $\omega((x, y))$ of H identifies H with its own Pontryagin dual. This implies that $Z/\operatorname{Ker}(\omega)$ is the center of $H^{\sim}/\operatorname{Ker}(\omega)$.

For every closed subgroup L of H, write $L^{\sim}$ for its inverse image in $H^{\sim}$. We shall say that L is totally isotropic if $\omega((x, y))$ is trivial on L, i.e. if $L^{\sim}/\operatorname{Ker}(\omega)$ is commutative.

We are interested in representations of $H^{\sim}$ with central character ω. Such a representation will be called admissible if, for every compact open totally isotropic subgroup K $\subset$ H, its restriction to $K^{\sim}$ is a sum of one-dimensional representations, each occurring with finite multiplicity.

A function f on $H^{\sim}$ will be called ω-locally constant if, for every x in $H^{\sim}$, there exist a K as above and a character χ of $K^{\sim}$ extending ω such that f(kx) = $\chi(k)f(x)$ for k $\in$ $K^{\sim}$.

Finally, fix a maximal totally isotropic subgroup L of H and a character ω' of $L^{\sim}$ extending ω. Under the preceding hypotheses it is known that :

(4.1.1) The group $H^{\sim}$ has an irreducible admissible representation V with central character ω. It is unique up to isomorphism.

(4.1.2) There is a nonzero linear form a on V such that $a(\ell v) = \omega'(\ell)a(v)$ for $\ell \in L^{\sim}$. It is unique up to a scalar.

(4.1.3) The map sending v to the function a(gv) on $H^{\sim}$ is an isomorphism of V with the space of ω-locally constant functions on $H^{\sim}$ satisfying $f(\ell g) = \omega'(\ell)f(g)$ for $\ell \in L^{\sim}$, and having compact support modulo $L^{\sim}$.

In particular, if H is finite, V has dimension equal to the index of a maximal totally isotropic subgroup of H, i.e. the square root of the order of H.

(4.2) Let F be a local field, and write simply $T^{\sim}$ for $T(F)^{\sim}$. By 1.8.1 and 2.2, the center of $T^{\sim}$ is the subgroup $\operatorname{diag}(F^{*n} \times F^{*n})^{\sim}$. The quotient of $T^{\sim}$ by its center is finite and commutative, and the commutator map has values in μ. The theory of Heisenberg groups gives the following.

Proposition 4.3 (i) The isomorphism class of an irreducible admissible ε-representation of $T^{\sim}$ is determined uniquely by its central character. (ii) Let ω be an ε-character of the center of $T^{\sim}$, let T' be a subgroup of T such that $T'^{\sim}$ is a maximal commutative subgroup of $T^{\sim}$, and let ω' extend ω to $T'^{\sim}$.

The induced representation $Ind_{T'^{\sim}}^{T^{\sim}}(\omega')$ is irreducible, with central character ω.

4.4 The center of $T^{\sim}$ is $(F^{*n} \times F^{*n})^{\sim}$. Its index in $T^{\sim}$ is $[F^* : F^{*n}]^2$, and the irreducible admissible ε-representations of $T^{\sim}$ therefore have dimension $[F^* : F^{*n}]$. The extension $(F^{*n} \times F^{*n})^{\sim}$ of $F^{*n} \times F^{*n}$ by μ is split by $diag^{\sim}$ (3.8). This identifies ε-characters ω of the center of $T^{\sim}$ with pairs (α_1, α_2) of characters of F^{*n}, by

(4.4.1) $\omega(diag^{\sim}(a, b)) = \alpha_1(a)\alpha_2(b)$.

We shall use this identification. By 4.3, the irreducible admissible ε-representations of $T^{\sim}$ are parametrized by these pairs (α_1, α_2). It is sometimes more convenient to parametrize them by pairs (χ_1, χ_2) of characters of F^* that are trivial on $\mu_n(F) \subset F^*$, by setting

(4.4.2) $\chi_i(x) = \alpha_i(x^n)$ (i = 1,2).

To realize the ε-representations of $T^{\sim}$, two choices of T' as in 4.3(ii) are convenient. In each case there is a natural splitting of the extension $T'^{\sim}$ of T' by μ, so giving an extension ω' of ω to $T'^{\sim}$ amounts to extending (α_1, α_2) from $F^{*n} \times F^{*n}$ to T'.

a) Subgroup $F^* \times F^{*n}$; splitting (a,b) $\mapsto$ $d_1(a)d_2(b)$. b) In the tame case : subgroup $O^*F^{*n} \times O^*F^{*n}$; splitting (a,b) $\mapsto$ $d_1(a)d_2(b)$.

In case (b), when χ_1 and χ_2 are unramified, there is a unique extension ω' of ω that is trivial on $d_1(O^*)d_2(O^*)$.

4.5 Now suppose F is global. The center of $T(A)^{\sim}$ is the inverse image of T(A) = diag$(A^{*n} \times A^{*n})$, and the inverse image of $T(F)T(A)^n$ is a maximal commutative subgroup. By 4.1 one has :

Theorem 4.6. Let α_1 and α_2 be two characters of A^{*n}/F^{*n}, and let V_{α_1,α_2} be the space of ε-functions on $[T(F)]\backslash T(A)^{\sim}$, right-invariant under an open subgroup of $T(A^f)^{\sim}$, and satisfying

$$f(g \cdot diag^{\sim}(a_1, a_2)) = f(g)\alpha_1(a_1)\alpha_2(a_2)$$
$$\text{if } a_1, a_2 \in A^{*n}.$$

Then V_{α_1,α_2} is an irreducible admissible ε-representation of $T(A)^{\sim}$.

These representations are the automorphic representations of T(A). It is sometimes more convenient to index them by pairs (χ_1, χ_2) of Hecke characters that are trivial on the local n-th roots of unity ; put $\chi_i(a) = \alpha_i(a^n)$.

5. The principal series

5.0. Let F be a local field. Write G for $\mathrm{GL}(2, F)$, and simplify the notation by omitting F : thus $T^{\sim}$ will stand for $T(F)^{\sim}$, and so on.

For F = C one has $G^{\sim}$ = G × μ, and the theory of representations of $G^{\sim}$ reduces to that of G. For F = R one has n = 1 or 2. We shall not dwell on these familiar cases : throughout the rest of this section, except in 5.12, F is assumed non-Archimedean.

5.1. Some arguments from the representation theory of reductive groups over F generalize unchanged to central extensions such as $\mathrm{SL}(2, F)^{\sim}$ or $\mathrm{GL}(2, F)^{\sim}$; others do not. Those that generalize are based on the Bruhat decomposition; the structure of the unipotent radical U of a parabolic P (root subgroups, ...); the Levi decomposition P = MU and the $contracting/dilating$ character of the action of M on U ; and the relation between the group and its Lie algebra. Examples include general results on Jacquet modules and supercuspidal representations. Difficulties arise from the commutativity of tori and the structure of Hecke algebras.

5.2. Write simply B and U for B^{+} and U^{+}. The group $B^{\sim}$ is the semidirect product $T^{\sim} \cdot U$ of $T^{\sim}$ by the lift (1.7) of U. If ρ is a representation of $T^{\sim}$, use the same symbol ρ for the representation of $B^{\sim}$ that is trivial on U and is given by $\rho(tu) = \rho(t)$.

The modulus of $B^{\sim}$ is thereby obtained from the quasi-character δ of $T^{\sim}$, defined by $\delta(t) = \|ab^{-1}\|$ when t lies above $\mathrm{diag}(a, b) \in$ T. If $\rho : T^{\sim} \to \mathrm{GL}(V)$ is an algebraic representation of $T^{\sim}$, the induced representation $Ind_{B^{\sim}}^{G^{\sim}}(V, \rho)$, or simply Ind(V), is the representation by right translation on $B^{\sim}\backslash G^{\sim}$ on the space of locally constant functions $f : G^{\sim} \to$ V satisfying

(5.2.1) f(utg) $= \delta(t)^{1/2}\rho(t)f(g)$.

5.3. Interpret this definition through the dictionary between :

(a) locally constant sheaves of complex vector spaces on P^{1}, equipped with an ("algebraic") action of $G^{\sim}$;

(b) algebraic representations (V, ρ) of $B^{\sim}$;

given by the equivalence of categories $\mathscr{F} \mapsto$ the fiber of $\mathscr{F}$ at the point (1,0) of P^{1}, whose stabilizer is $B^{\sim}$.

If to a section f of $\mathscr{F}$ one associates the function f(g) equal to the germ at (1,0) of $g^{-1}f$, then $\Gamma(\mathscr{F})$ is identified with the space of locally constant functions $f : G^{\sim} \to$ V satisfying f(bg) $= \rho(b)f(g)$. For V = C and the representation given by the character ut $\mapsto \delta(t)$, one obtains the sheaf of locally constant measures, i.e. locally of the form $\|\omega\|$ for a differential form ω. Finally, the space of functions 2.2.1 is interpreted as the space of locally constant densities of weight $1/2$ with values in the equivariant sheaf defined by V.

5.4. Let us spell out, for the group $G^{\sim}$, what the general theory of Jacquet modules gives. For a very clear account, see [1] (it discusses GL(n), but the arguments are general).

(A) On the category of algebraic representations of U, the coinvariant functor W $\mapsto$ W_U is exact.

This exactness makes it easy, up to an extension, to calculate $Ind(V,\rho)_U$. Write proj for the projection to coinvariants. As in 2.3, write $Ind(V,\rho) = H^0(P^1, \mathscr{F})$ for a suitable equivariant sheaf. The exact sequence

$$0 \to H^0_c(P^1 - (1,0), \mathscr{F}) \to H^0(P^1, \mathscr{F}) \to \mathscr{F}_{(1,0)} \to 0$$

is $B^{\sim}$-equivariant. Passing to coinvariants gives an analogous exact sequence. In the quotient, $\mathscr{F}_{(1,0)} \simeq \mathscr{F}_{(1,0),U}$, identified with V by proj f $\mapsto$ f(e). In the subobject, $H^0_c(P^1 - (1,0), \mathscr{F})_U$ is again identified with V by integration over U (of which $P^1 - (1,0)$ is a principal homogeneous space). The functoriality will be made explicit below.

(B) The induction functor from algebraic representations of $T^{\sim}$ to those of $G^{\sim}$ has as left adjoint the functor r sending W to the coinvariant quotient W_U, with $T^{\sim}$ acting by

$$t \cdot proj(w) = \delta(t)^{-1/2} proj(tw).$$

After choosing w $\in$ $N(T^{\sim}) - T^{\sim}$ and a Haar measure on U, calculation (A) gives an exact sequence of representations of $T^{\sim}$:

(5.4.1) $0 \to (V, \rho \circ int_w) \to$ r $Ind(V,\rho) \to (V,\rho) \to 0$,

where the projection to (V,ρ) is the adjunction map proj(f) $\mapsto$ f(e), and, when f(e) = 0, the image of proj(f) in $(V, \rho \circ int_w)$ is

(5.4.2) proj(f) $\mapsto$ $\int$ f(wu)du (for f(e) = 0).

The condition f(e) = 0 ensures that the integrand has compact support.

(C) Let (V,ρ) be an irreducible admissible representation of $T^{\sim}$. Then every nonzero subquotient W of $Ind(V,\rho)$ satisfies rW $\neq$ 0.

The idea of the proof (cf. [2], 2.9) is that if rW = 0, then W is projective, hence also a subrepresentation of $Ind(V,\rho)$, contradicting :

(D) For (V, ρ) as above, and any nonzero subrepresentation W $\subset$ $Ind(V, \rho)$, the map rW $\to$ (V, ρ) is onto. Otherwise the f $\in$ W would all be identically zero !

5.5. Let (V, ρ) be an irreducible admissible ε-representation of $T^{\sim}$, with central character (α_1, α_2) $(4.3, 4.4)$.

If $\alpha_1 \neq \alpha_2$, the representations ρ and $\rho \circ int_w$ have distinct central characters, and the sequence 5.4.1 splits uniquely. The splitting

(5.5.1) r $Ind(V, \rho) \to (V, \rho \circ int_w)$

regularizes the integral 5.4.2 for f not necessarily satisfying f(e) = 0. For $\alpha_1(a) \neq \alpha_2(a)$, it is given by

(5.5.2) f $\mapsto$ $(1 - \alpha_1(a)^{-1}\alpha_2(a))^{-1} \int f_a(wu)du$,

with

$$f_a(g) = \text{f(g)} - \alpha_1(a)^{-1}\|a\|^{-1/2} f(gd_1(a))$$
(which vanishes at g = e).

By adjunction, (5.5.1) gives a morphism

(5.5.3) $I : Ind(V, \rho) \to Ind(V, \rho \circ int_w)$,

given by the intertwining integral f $\mapsto$ $\int f(wug)du$, regularized by (5.5.2). Note that 5.5.2 gives an unconditional definition (even when $\alpha_1 = \alpha_2$) of "$(1 - \alpha_1(a)^{-1}\alpha_2(a))I$". One checks :

Lemma 5.6. If $\alpha_1 = \alpha_2$ and a has nonzero valuation, then

(i) r $Ind(V, \rho)$ is an indecomposable representation of $T^{\sim}$;

(ii) the expression "$(1 - \alpha_1(a)^{-1}\alpha_2(a))I$" above is a nonzero scalar multiple of the identity.

5.7. Let (V, ρ) and (α_1, α_2) be as in 5.5. By 5.4(A),(C), the functor r embeds the lattice of subrepresentations of $Ind(V, \rho)$ into that of r $Ind(V, \rho)$; and by (D), if W is a proper subrepresentation, then rW defines a splitting of the exact sequence 5.4.1. Hence :

a) if $\alpha_1 = \alpha_2$, 5.6 gives the irreducibility of $Ind(V, \rho)$;

b) if $\alpha_1 \neq \alpha_2$, $Ind(V, \rho)$ has at most one proper subrepresentation, and is therefore indecomposable ;

c) if W is a proper subrepresentation of $Ind(V, \rho)$, then rW is the kernel of $5.5.1$, and W lies in the kernel of the intertwining operator I. The $T^{\sim}$-morphism rI annihilates rW $\simeq (V, \rho)$, and its image is a splitting of $(5.4.1)$ for $(V, \rho \circ int_w)$. Thus the image of I is a proper subrepresentation of $Ind(V, \rho \circ int_w)$.

If α is a character of F^*, write $\alpha \circ det$ for the character of $G^{\sim}$ given by $G^{\sim} \to \mathrm{GL}(2, F) \to^{det} F^*$. One has $Ind(V, \rho) \otimes (\alpha \circ det) = Ind(V, \rho \cdot \alpha \circ det)$, and the central character of $\rho \cdot \alpha \circ det$ is $(\alpha_1 \alpha, \alpha_2 \alpha)$. Such a twist does not change whether $Ind(V, \rho)$ is reducible or irreducible. If $\sigma = \alpha_1 \alpha_2^{-1}$ is unitary, such a twist can make (V, ρ), and hence $Ind(V, \rho)$, unitary. Altogether :

Proposition 5.8 (i) The representation $Ind(V, \rho)$ is indecomposable. If $\sigma = \alpha_1 \alpha_2^{-1}$ is unitary, it is irreducible.

(ii) If $\sigma \neq 1$, one of the following holds :

(a) $Ind(V, \rho)$ is irreducible, and I is an isomorphism ;

(b) $Ind(V, \rho)$ and $Ind(V, \rho \circ int_w)$ are reducible, and I induces an isomorphism from the unique nontrivial quotient of $Ind(V, \rho)$ to the unique nontrivial subrepresentation of $Ind(V, \rho \circ int_w)$.

5.9. The Jacquet module of a representation controls the asymptotic behavior of its matrix coefficients. It follows easily that, when $Ind(V, \rho)$ is reducible, exactly one of the two representations of which it is an extension has square-integrable coefficients (i.e. belongs to the discrete series). It will be called special. The irreducible representations $Ind(V, \rho)$, together with the nonspecial constituents of the reducible $Ind(V, \rho)$, form the principal series.

5.10. For Re $\sigma > 1$ (notation $0.0.7$), the intertwining integral converges. Standard procedures make it possible to treat σ as a parameter ; then $5.5.2$ gives the analytic continuation of this integral. An elegant way to proceed was pointed out to me by Bernstein. It consists in considering representations with coefficients not in C, but in an algebra such as $C[T, T^{-1}]$. Its advantage over ordinary analytic continuation is that it shows the intertwining integral I to depend rationally on σ, not merely meromorphically. The regularization formula $5.5.2$ shows that, on putting $L(\sigma) = 1$ when $\sigma(x^n)$ is ramified, and $L(\sigma) = (1 - \sigma(\pi^n))^{-1}$ when σ is unramified and π is a uniformizer, the operator $I/L(\sigma)$ even depends polynomially on σ.

Iterating the intertwining integral gives an endomorphism of $Ind(V, \rho)$. When $Ind(V, \rho)$ is irreducible, this is multiplication by a nonzero scalar.

When $Ind(V, \rho)$ is reducible, the iterated operator is again a scalar (by analytic continuation); it is therefore 0.

5.10. C. Moen determined when $Ind(V, \rho)$ is reducible. I prefer to state his result in terms of the characters χ_i of F^*, defined by $\chi_i(x) = \alpha_i(x^n)$.

Theorem 5.11 (C. Moen). The representation $Ind(V, \rho)$ is irreducible except when

$$\chi_1 \chi_2^{-1} = \|x\|^{\pm 1}.$$

When $\chi_1 \chi_2^{-1} = \|x\|$, it is the subrepresentation of $Ind(V, \rho)$ that belongs to the discrete series. When $\chi_1 \chi_2^{-1} = \|x\|^{-1}$, it is the quotient representation.

5.12. This theorem determines the complementary series. If $Ind(V, \rho)$ is unitary, it is isomorphic to the contragredient of its complex conjugate, and either χ_1 and χ_2 are unitary (the principal series in the strict sense), or $\chi_2 = \overline{\chi}_1^{-1}$. A twist (5.7) reduces the latter case to $\chi_1 = \|x\|^s$ and $\chi_2 = \|x\|^{-s}$, with s ∈ R. The corresponding induced representation π_s is real, with contragredient π_{-s}, and the intertwining operator $I/L(\sigma)$ supplies an invariant bilinear form on π_s. For s = 0 it is positive definite. By analytic continuation it remains so as long as $I/L(\sigma)$ is invertible, i.e. for $|s| < 1$. At s = 1 the two constituents of π_s are unitary (take s = 1 for one and s = -1 for the other). Beyond this, π_s cannot be unitary; otherwise it would remain so as s → ∞, which is impossible, since the coefficients are unbounded for large s.

5.13. Y. Flicker [3] reduced the classification of ε-representations of $G^\sim$ to the classification of representations of G. He establishes a bijection, characterized in terms of characters, between isomorphism classes of irreducible admissible ε-representations of $G^\sim$ and isomorphism classes of irreducible admissible representations π of G satisfying :

a) if π is in the principal series, $\pi = \pi(\chi_1, \chi_2)$, and χ_1 and χ_2 are trivial on $\mu_n(F) \subset F^*$;

b) if π is in the discrete series, its central character ω_π is trivial on $\mu_n(F) \subset F^*$.

If $\pi^\sim$ corresponds to π, the central character of $\pi^\sim$ (a character of F^{*n}) is $\omega_\pi(x^n)$. If $\pi^\sim$ is in the principal series defined by the representation of $T^\sim$ with central character (α_1, α_2), and $\chi_i = \alpha_i(x^n)$, then π is $\pi(\chi_1, \chi_2)$. In particular, if $(\chi_1, \chi_2) = (\|x\|^{1/2}, \|x\|^{-1/2})$, the irreducible quotient of $Ind(V, \rho)$ corresponds to the trivial representation of G.

5.14. Let ψ be a nontrivial character of U, and repeat the calculation in 5.7 of $(Ind(V,\rho))_U$ in order to calculate $(Ind(V,\rho)\otimes\overline{\psi})_U$. With the notation of 5.7, $\mathscr{F}_{(1,0)}$ contributes nothing, since the action of U there is trivial. Integration of the term $H_c^0(P^1-(1,0),\mathscr{F})$ again gives a copy of V :

Proposition 5.15. There exists a unique linear map

$$\varphi : Ind(V,\rho) \to \text{V}$$

given by $\varphi(f)=\int f(wu)\psi(u)du$ for f(e) = 0 and satisfying $\varphi(uf)=\psi(u)\varphi(f)$. It identifies V with $(Ind(V,\rho)\otimes\overline{\psi})_U$.

If $\psi(u_1)\neq 1$, one also has

(5.15.1) $\varphi(f)=(1-\psi(u_1))^{-1}\int(f(wu)-f(wuu_1))\overline{\psi}(u)du.$

As in 5.10, this formula may be regarded as an analytic continuation in σ of the integral $\int f(wu)du$, which converges for Re $\sigma>1$.

5.16 For a representation W of $G^{\sim}$, one has on (W $\otimes\,\overline{\psi})_U$, if not an action of $T^{\sim}$, at least an action of the centralizer $Z^{\sim}$ of ψ in $T^{\sim}$; define τ_ψ W to be the representation of $Z^{\sim}$ on (W $\otimes\,\overline{\psi})_U$ given by $z\cdot proj(f)=proj(z\cdot f)$. Since the character δ is trivial on Z, the morphism $\varphi:\tau_\psi\ Ind(V,\rho)\to$ V of 5.15 is a morphism of representations of $Z^{\sim}$.

5.17 As in 4.2, one checks that the irreducible ε-representations of $Z^{\sim}$ are classified by their central character. If n is odd (respectively, if n = 2m), the center of $Z^{\sim}$ is the inverse image of the subgroup F^{*n} (respectively F^{*m}) of the center F^* of G. As a representation of $Z^{\sim}$, V

(a) for odd n, is a sum of $[F^*:F^{*n}]^{1/2}$ irreducible representations, all isomorphic, with central character induced by ω_ρ;

(b) for even n = 2m, has as its irreducible subrepresentations the irreducible representations of $Z^{\sim}$ whose central character extends ω_ρ (there are $[F^{*m}:F^{*n}]$ nonisomorphic ones). Each occurs with multiplicity $[F^*:F^{*m}]^{1/2}$.

5.18. To calculate the functor r_ψ on the subquotients of the principal series, it suffices, by 5.8, to calculate the morphism induced by the intertwining integral—

continued

(5.18.1) r_ψ I : $r_\psi\ Ind(V, \rho) \to\ r_\psi\ Ind(V, \rho \circ int_w)$

for ρ with central character ω_ρ defined by (χ_1, χ_2), where $\chi_1 \chi_2^{-1} = \|x\|$. By 2.14, this $Z^{\sim}$-morphism is identified with an endomorphism of V.

Theorem 5.19. Let ρ be a representation of $T^{\sim}$ with central character $(\alpha_1, \alpha_2) = (\alpha \cdot \|x\|^{1/n}, \alpha)$. Putting $\chi_i = \alpha_i(x^n)$, one has $(\chi_1, \chi_2) = (\chi \cdot \|x\|, \chi)$. Let W be the irreducible quotient of $Ind(\rho)$. In the tame case, for n odd, the representation $(W \otimes \overline{\psi})_U$ of $Z^{\sim}$ is a sum of $(n-1)/2$ mutually isomorphic representations.

When n = 3, $(n-1)/2 = 1$; this uniqueness result explains the "multiplicative" appearance of the statements in the Introduction.

6. Models, and proofs.

6.1 Let α_1 and α_2 be two characters of F^{*n}. We shall describe an irreducible ε-representation ρ of $T^{\sim}$ with central character (α_1, α_2), and describe the induced representation $Ind(\rho)$ as a space of functions on the cover P of $F^2 - 0$. It is the space of locally constant ε^{-1}-functions $(f(\zeta p) = \varepsilon(\zeta)^{-1} f(p))$ having a suitable homogeneity with respect to the "homotheties" v $\mapsto$ λv for $\lambda \in F^{*n}$ (3.8).

Choose an extension, still denoted α_2, of α_2 to F^*, and describe ρ by 4.4(a). The representation $Ind(\rho)$ is then obtained by induction in stages : from $d_1(F^{*n})d_2(F^*)\mu$ to $T^{\sim}$, and then from $B^{\sim}$ to $G^{\sim}$, starting with (α_1, α_2). It is the space of ε-functions f on $G^{\sim}$ such that

$$f(u\, d_1(a) d_2(b) g) = \alpha_1(a)\alpha_2(b)\|a/b\|^{1/2} f(g)$$

for u $\in$ U, a $\in$ F^{*n}, and b $\in$ F^*. If the character $\alpha_2(b)\|b\|^{-1/2}$ is trivial, this is identified with a space of functions on $d_2(F^*) \cdot U \backslash \mathrm{GL}(2, F)^{\sim}$. This quotient is precisely the cover P of $F^2 - 0$: associate to g $\in$ $\mathrm{GL}(2, F)^{\sim}$ the point $g^{-1}((1, 0), e_X)$. Reducing to this case by twisting by a determinant character,

det : $\mathrm{GL}(2, F)^{\sim} \to \mathrm{GL}(2, F) \to F^*$,

one obtains the following model.

a) The representation space consists of locally constant ε^{-1}-functions on P with homogeneity $(\alpha_2 - \alpha_1 - 1)$:

$$f(\lambda v) = (\alpha_2 - \alpha_1 - 1)(\lambda) f(v) \text{ for } \lambda \in F^{*n}, \text{ in the notation of } 0.0.7.$$

b) The group acts by

$$(g*f)(v) = f(g^{-1}v) \cdot (\alpha_2 - 1/2)(det\ g).$$

6.2 Another model of the same representation is obtained from the dual plane. Choose this time an extension of α_1 to F^*, and take

a) as space, the locally constant ε^{-1}-functions f on $P^{\vee}$ satisfying

$$f(\lambda v) = (\alpha_2 - \alpha_1 - 1)(\lambda) f(v) \text{ for } \lambda \in F^{*n} \ (0.7, 3.9);$$

b) as action,

$$(g * \varphi)(v) = \varphi(g^{-1}v)(\alpha_1 + 1/2)(det\ g).$$

6.3 In these models, the intertwining operator from the induced representation with parameters (α_1, α_2) to that with parameters (α_2, α_1) is a Radon transform. To f on P, of homogeneity $(\alpha_2 - \alpha_1 - 1)$, it associates Rf on $P^\vee$, of homogeneity $(\alpha_1 - \alpha_2 - 1)$. The operator R is as follows. A point $\ell \in P^\vee$ determines

a) a point $\overline{\ell}$ in $A^{2\vee} - 0$, and a line $\overline{\ell}(\overline{v}) = 1$ in A^2;

b) a trivialization of P over this line. We write $< \ell, v > = 1$ for the relation that $\overline{\ell}(\overline{v}) = 1$ and v lies in the section of P over the line $\overline{\ell}(\overline{v}) = 1$ determined by ℓ.

Set

$$(Rf)(\ell) = \int_{<\ell,v>=1} \mathrm{f(v)}\ (dxdy/d\ell).$$

This formula presupposes the choice of a Haar measure on F. With a pardonable abuse of notation, it may also be written

$$(Rf)(\ell) = \iint \mathrm{f(v)}\ \delta(< \ell, v > -1)\ \mathrm{dxdy}.$$

The same formula defines the Radon transform from functions on $P^\vee$ to functions on P; this is again a model of the intertwining operator.

The regularization $(5.5.2)$ of the integral is the usual “finite part,” except when $\alpha_1 = \alpha_2$, the case corresponding to the pole of the intertwining operator.

6.4 Let us determine the map Φ of 5.15. Write f(x,y ;p) for a function on P, where p $\in$ P lies above (x,y). For an ε^{-1}-function f on P, of homogeneity $(\alpha_2 - \alpha_1 - 1)$, as in 6.2, put

$$\mathrm{L(f;y)} = \int f(x, y; e_y)\ \psi(x/y)\ \mathrm{dx}.$$

For large x, and for λ in a suitable open subgroup of O^{*n}, one has $f(x, y; e_y) = f(\lambda x, y; e_y)$. This permits the regularization of the integral as in $5.15.1$. One has

(6.4.1) $L(e^+(u)f; y) = \psi(u)L(f; y)$, and

(6.4.2) $L(f; \lambda y) = (\alpha_2 - \alpha_1)(\lambda)L(f; y)$ for $\lambda \in F^{*n}$,

(6.4.3) $L(d_1(z)d_2(z)f; y) = \|z\|\varepsilon(z, y)^{-1}L(f; z^{-1}y)$, for z $\in F^*$.

The space of functionals satisfying $6.4.1$ has as a basis the L(f ;y), with y running over F^*/F^{*n}. These are the coordinates of Φ (5.15) in a suitable basis.

6.5 **In the dual-plane model, the coordinates of Φ are likewise**

$$\mathrm{L(f;a)} = \int f(a, b; e_a)\ \overline{\psi}(b/a)\ \mathrm{db}.$$

One has

(6.5.1) $L(e^{+}(u)f;a) = \psi(u)L(f;a)$,

(6.5.2) $L(f;\lambda a) = (\alpha_2 - \alpha_1)(\lambda)L(f;a)$ for $\lambda \in F^{*n}$,

(6.5.3) $L(d_1(z)d_2(z)f;a) = \|z\|\varepsilon(z,z^{-1}a)^{-1}L(f;z^{-1}a)$.

The action in 5.18 of the intertwining operator is given by

(6.5.4) L(Rf ;a) = $\int \varepsilon(a,y)\,\overline{\psi}(1/ay)L(f;y)d^{*}y$,

$$= \sum_{y \in F^{*}/F^{*n}} L(f;y)\varepsilon(a,y)$$

$$\cdot \int_{F^{*n}} \overline{\psi}(\lambda/ay)(\alpha_1-\alpha_2)(\lambda)d^{*}\lambda,$$

where the integral is defined by analytic continuation in $(\alpha_1 - \alpha_2)$. There is a pole at $\alpha_1 = \alpha_2$, as required.

6.6 By 5.18 this reduces the proof of 5.19 to the following problem. Consider

V' : the space of locally constant functions L(y) on F^{*} satisfying (6.4.2), on which $Z^{\sim}$ acts by the right-hand side of 6.4.3;

V'' : the space of locally constant functions L(a) on F^{*} satisfying (6.5.2), on which $Z^{\sim}$ acts by the right-hand side of 6.5.3;

R : V' $\to$ V'', given by 6.5.4 :

$$\text{(RL)(a)} = \sum_{y \in F^{*}/F^{*n}} L(y)\varepsilon(a,y)$$

$$\cdot \int_{F^{*}} \overline{\psi}(\lambda/ay)(\alpha_1-\alpha_2)(\lambda)d^{*}\lambda\,;$$

and seek the representation Im R of $Z^{\sim}$ when R is not an isomorphism, i.e. when the character $(\alpha_1 - \alpha_2)$ is $\|\lambda\|^{\pm 1/n}$.

6.7 Under the hypotheses of 5.19, it is enough to restrict R to the invariants of $d_1(a)d_2(a)|a \in O^{*}$. Using as bases the functions L(y) or L(a) supported on a coset of $O^{*} \cdot F^{*n}$, R is represented by a matrix that decomposes along the diagonal into one 1×1 block and $(n-1)/2$ blocks of size 2×2 (in a suitable ordering). The calculation is then easy.

7. Eisenstein Series.

7.1 The "Eisenstein-series" operator E is an intertwining operator from

$$Ind_{B^{*}(A)^{\sim}}^{G(A)^{\sim}}(V_{\alpha_1,\alpha_2})$$

into the space of functions on $G(F)\backslash G(A)^{\sim}$. The space of the induced representation is identified with the space of ε-functions on $U(A)T(F)\backslash G(A)^{\sim}$, right-invariant under an open compact subgroup of $G(A^f)^{\sim}$ and K_∞-finite, and satisfying

$$f(diag^{\sim}(a_1,a_2)g) = \alpha_1(a_1)\alpha_2(a_2)\|a_1a_2^{-1}\|^{1/2}\text{ f(g)}$$
$$(a_1,a_2 \in A^{*n}).$$

The operator is

$$\sum_{\gamma \in B(F)\backslash G(F)} f(\gamma g),$$

when this converges, and in general is obtained by analytic continuation in α_1, α_2. Put $\chi(x) = \alpha_1(x^n)\alpha_2(x^n)^{-1}$. Classical arguments ([11]) produce a pole at $\chi = \|x\|$. If Ē is the residue of E, Ē intertwines the irreducible quotient of $Ind_{B^{*}(A)^{\sim}}^{G(A)^{\sim}}(V_{\alpha_1,\alpha_2})$, for $\chi = \|x\|$, into the space of functions on $G(F)\backslash G(A)^{\sim}$.

7.2 This determines the isomorphism class of the representation of $G(A)^{\sim}$ that is the image of Ē. Let ψ be an additive character of A/F. The "Fourier coefficient" operator

f $\mapsto$ W(f;g) = $\int_{U(F)\backslash U(A)}$ f(ug) $\overline{\psi}(u)$ du

commutes with right translations and satisfies W(f;ug) = $\psi(u)W(f;g)$. One also has W(f;zg) = W(f;g) for z $\in$ Z(F).

7.3 For n = 3, these properties and the uniqueness results of 5.19 should make it possible to identify it - and to obtain the results announced in the Introduction. Another method consists in changing tack and systematically replacing GL$(2,F)$, GL$(2,F_v)$, or GL$(2,A)$ by the subgroup of elements whose determinant is a cube. For this new group, the inverse image of the center is commutative, which simplifies the uniqueness results : the spaces $V_{U,\psi}$, previously irreducible representations of the inverse image of the center, become one-dimensional. Put

$f^0(g) = \int_{U(F)\backslash U(A)}$ f(ug) du (constant term).

One also has

$$f(g) = f^0(g) + \sum_{\lambda \in F^*} W(f;\ \mathrm{diag}(\lambda^2, \lambda)g).$$

If f, as a vector in the representation of $G(A)^{\sim}$ in the image of Ē, is decomposable, the function on $A^{f*\sim}$, $W(f; d_1(a^2)d_2(a))$, is multiplicative in the sense of the Introduction, from which the results follow.

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